

KYLE KAMAY

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EDUCATION

University of Connecticut, Storrs, CT

Honors Bachelor of Science, May 2029

Major: Electrical Engineering; GPA: 3.86

Electrical Engineering Experience

UConn Tech Park – Travelers Project Challenge, Hartford, Connecticut

Engineering Project Participant, May 2026 - September 2026

- Developed a Python-based physics and statistical model to estimate hail damage probability across multiple material types using impact energy, material properties, and inspection data
- Implemented Monte Carlo simulations to model randomized storm conditions and compare predicted damage risk across roofing, automotive, and structural materials
- Conducted technical and competitive research on weather patterns, hail dynamics, drone technology, and industry approaches to improve property-damage assessment
- Collaborated with Travelers professionals to analyze existing damage-assessment processes and developed a web-based platform to communicate research findings, technical analysis, and recommendations

UConn Nursing & Engineering Innovation Center, Storrs, Connecticut

NurseEng Fellow, August 2025 - May 2026

- Designed and prototyped a pressure-sensing system using FSR sensors and an ESP32-based microcontroller to collect and analyze real-time sensor data
- Developed embedded hardware and data-acquisition methods while iterating on sensor placement and system architecture
- Collaborated with nursing students and faculty to turn clinical needs into technical requirements and iterate on a functional prototype

Engineering Projects

Embedded Patient Pressure Monitoring System

- Integrated FSR sensors, FireBeetle ESP32, analog multiplexing, and Bluetooth communication to acquire and transmit data
- Programmed embedded systems to collect, process, and visualize sensor data while evaluating hardware architecture

Low-Cost Heart Rate Monitoring System

- Designed and prototyped a wearable-style heart rate monitoring system using a pulse sensor and Arduino Uno for real-time physiological data acquisition
- Programmed Arduino/C++ software to collect, process, visualize, and interpret heart rate measurements

Leadership & Activities

IEEE – Member

- Applied circuit-analysis concepts to hands-on microcontroller projects, developing experience with electronic components, sensors, and embedded systems

UConn Underwater Robotics – Member

- Contributing to robot software development, programming systems for movement, visual perception, and autonomous/remote operation

UConn Formula One – EV – Contributor

- Helped team place 5th overall by assisting with electrical harness development and gained hands-on experience with electrical systems used in electric vehicles

Biomedical Engineering Society – Member

- Engaging with biomedical engineering opportunities, technical events, and professional development activities to broaden engineering knowledge

TECHNICAL SKILLS

Programming: Python, C++

Embedded Systems: Arduino, ESP32, Sensor Integration, Data Acquisition, Circuit Analysis, Breadboarding

Software: GitHub, Microsoft Office, LTspice, PyCharm, Visual Studio Code